

Politecnico di Torino

POLITECNICO DI TORINO

Corso di laurea in
Ingegneria Matematica

Bioquants assignment
Breast cancer gene expression analysis

Giuliana Paradiso s319688

1 Introduction

Breast cancer represents a major public health challenge. Understanding the molecular mechanisms underlying breast cancer development is essential for improving diagnosis and treatment. Gene expression profiling technologies provide a powerful tool to investigate differences between tumor and normal tissues.
In this study, a publicly available breast cancer gene expression dataset was analyzed to identify genes that are differentially expressed between breast cancer samples and normal control samples. The dataset was obtained from the Gene Expression Omnibus (GEO) and consists of microarray-based expression measurements for a large number of genes across multiple samples. Each sample represents a biological tissue specimen, while each feature corresponds to a gene expression probe.

An end-to-end analysis pipeline was implemented using Python. The workflow included data import and preprocessing, quality control checks for missing values and outlier samples, differential expression analysis using statistical hypothesis testing, correction for multiple testing, and functional enrichment analysis. Differentially expressed genes were identified using a Welch's t-test, followed by false discovery rate (FDR) correction using the Benjamini–Hochberg procedure.
Finally, Gene Ontology (GO) and KEGG pathway enrichment analyses were performed to identify biological processes and pathways overrepresented among the differentially expressed genes.
This analysis aims to highlight key differences between breast cancer and normal samples and to provide biological insights into pathways potentially involved in breast cancer development.

2 Dataset

The dataset used in this study was obtained from the Gene Expression Omnibus (GEO) GSE15852, which contains gene expression measurements for both breast cancer and normal samples. The data were imported into Python using the `pandas` library for subsequent analysis. The processed Series Matrix file was read into a `DataFrame` and transposed so that rows corresponded to individual samples and columns to gene expression probes, resulting in a dataset of 86 samples and 22,283 probe-level expression measurements.

Sample labels distinguishing tumor and normal tissues were extracted from the `Sample_source_name_ch1` metadata field. Samples annotated as "breast tumor tissue" were classified as tumor, while those annotated as "normal breast tissue" were classified as controls.

3 Quality Control Checks

Before performing differential expression analysis, it is essential to assess the quality of the dataset to ensure the reliability of the results. Missing values and outliers can introduce bias and distort statistical analyses if not properly addressed. In the present dataset, no missing values were detected, as verified using the `pandas isnull` function.

Consequently, no samples were removed due to missing data.
Outlier detection was performed using boxplots and z-scores calculated across all genes for each sample. None of the samples exceeded the z-score threshold of 3, indicating the absence of extreme outliers, and therefore all samples were retained for subsequent analyses.

4 Statistical analysis

Differential gene expression analysis was performed using an independent two-sample t-test (Welch's t-test) to compare breast cancer samples and normal controls for each gene. P-values were subsequently adjusted for multiple testing using the Benjamini–Hochberg false discovery rate (FDR) procedure. Genes with an adjusted p-value below 0.05 were considered significantly differentially expressed. Using this criterion, **a total of 1,667 genes were identified as differentially expressed.**

5 Functional enrichment analysis

Since the microarray platform reports probe-level expression, probe identifiers were mapped to gene symbols using the official annotation file for the Affymetrix Human Genome U133A Array (GPL96), obtained from the GEO platform database (GPL96) prior to functional enrichment analysis. Out of 1,667 differentially expressed probes, 1,398 unique gene symbols were successfully mapped and used for functional enrichment. Some probes map to multiple genes; in such cases all mapped genes were considered for enrichment analysis.

To obtain a comprehensive interpretation of the results, both Gene Ontology and KEGG pathway enrichment analyses were performed. GO analysis was used to identify enriched biological processes, while KEGG analysis provided insight into the molecular pathways involved in breast cancer. Enrichment p-values for GO and KEGG terms were calculated using a hypergeometric test, and adjusted for multiple testing using the Benjamini–Hochberg procedure, as implemented in Enrichr.

Gene Ontology enrichment analysis identified **255 significantly enriched biological processes** (FDR-adjusted p-value < 0.05). Table 1 reports the top 10 GO Biological Processes ranked by adjusted p-value. KEGG pathway enrichment analysis revealed **44 significantly enriched pathways** (FDR-adjusted p-value < 0.05). The top 10 pathways are summarized in Table 2.

GO Term	Overlap	Adjusted P-value	Representative Genes
negative regulation of cellular process	81/566	2.0e-6	APP, BTG1, PLXND1, FHL1, PDE3B ...
regulation of angiogenesis	40/203	4.0e-6	SP100, SPARC, BTG1, FLT1, PLXND1 ...
negative regulation of apoptotic process	70/485	7.0e-6	CITED2, APIP, FHL2, AQP1, MDK ...
regulation of cell population proliferation	97/764	7.0e-6	APP, CD81, IRS2, CLU, FGF2 ...
regulation of apoptotic process	93/742	2.3e-5	TOP2A, FOXA1, CITED2, ITSN1, APIP ...
negative regulation of programmed cell death	57/381	3.1e-5	CITED2, APIP, FHL2, AQP1, SOX9 ...
negative regulation of blood vessel morphogenesis	21/78	3.6e-5	SPARC, NPR1, STAT1, SERPINF1, PDE3B ...
regulation of fat cell differentiation	21/80	4.6e-5	CDS1, CEBPB, XBP1, LMO3, RARRES2 ...
negative regulation of angiogenesis	22/87	4.6e-5	SPARC, NPR1, STAT1, SERPINF1, PDE3B ...
extracellular structure organization	38/216	5.4e-5	APP, COL15A1, SPARC, ECM2, LAMA2 ...

Table 1: Top 10 significant GO Biological Processes enriched in breast cancer differentially expressed genes.

KEGG Pathway	Overlap	Adjusted P-value	Representative Genes
PPAR signaling pathway	22/74	1.0e-6	GK, ACSL1, ADIPOQ, LPL, NR1H3 ...
Proteoglycans in cancer	37/205	1.0e-5	SDC4, SDC2, ITGB3, HSPB2, TWIST1 ...
Fatty acid degradation	15/43	1.0e-5	ACAA2, ECHS1, ADH1C, ADH1B, ACSL1 ...
Propanoate metabolism	13/34	1.6e-5	ECHS1, ABAT, ACACB, ACAT1, LDHB ...
Pyruvate metabolism	15/47	2.4e-5	ADH1C, MDH1, ADH1B, GLO1, ACACB ...
Focal adhesion	34/201	7.0e-5	FLT1, LAMA2, ITGB4, ITGB3, LAMA4 ...
ECM-receptor interaction	20/88	9.3e-5	VWF, LAMA2, SDC4, ITGB4, LAMB2 ...
Mitophagy	16/68	5.3e-4	BNIP3L, GABARAPL1, JUN, CSNK2A1, CITED2 ...
Valine, leucine and isoleucine degradation	13/48	5.9e-4	ACAA2, ECHS1, MCCC1, ABAT, ACAT1 ...
Citrate cycle (TCA cycle)	10/30	6.9e-4	CS, SUCLA2, MDH1, IDH1, IDH2 ...

Table 2: Top 10 significant KEGG pathways enriched in breast cancer differentially expressed genes.

6 Visualization of Results

A volcano plot was used to visualize differential gene expression between breast cancer and normal samples. The x-axis shows the t-statistic from Welch's t-test, while the y-axis represents the $-\log_{10}$ Benjamini–Hochberg adjusted p-values. Genes above the significance threshold (FDR < 0.05) are highlighted in red.

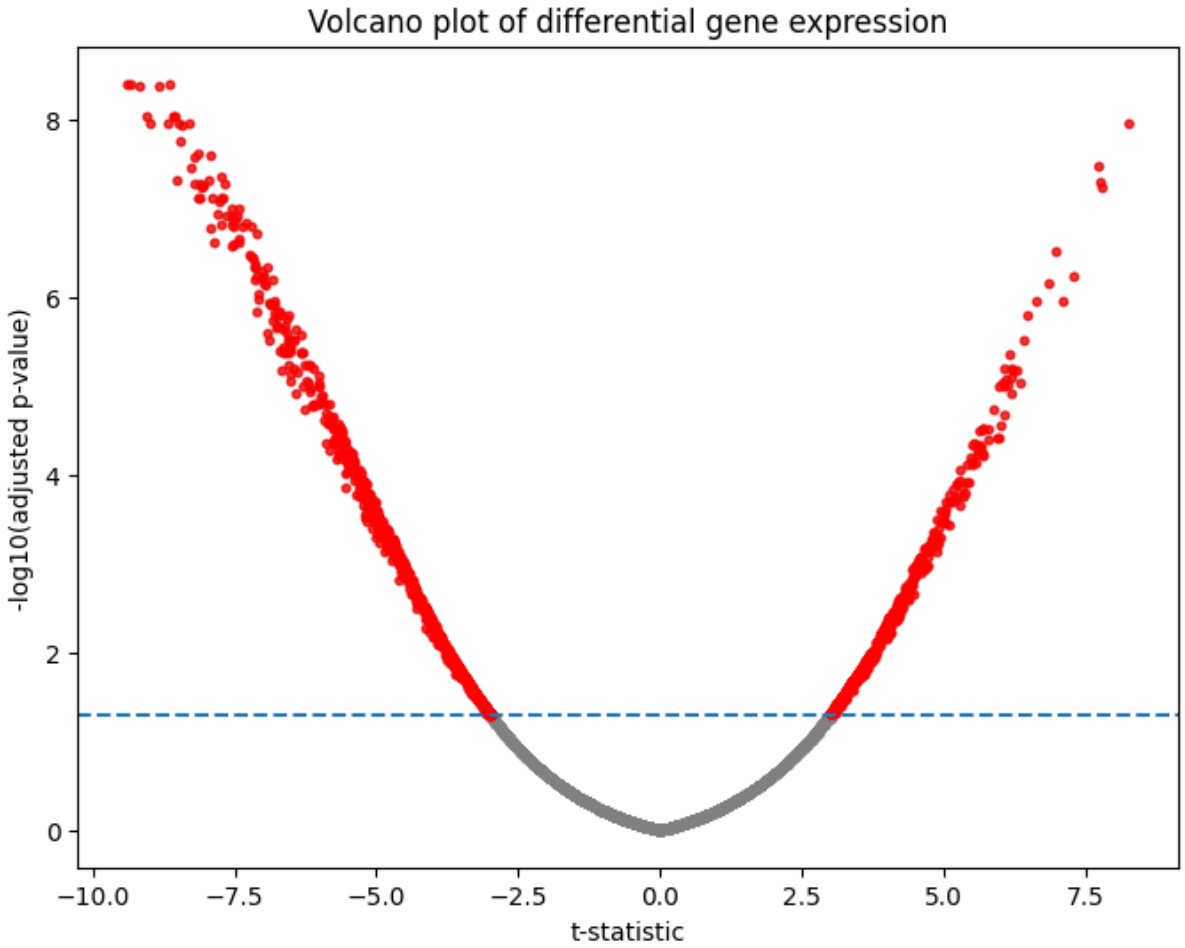


Figure 1: Volcano plot of differential gene expression between breast cancer and normal samples.

A heatmap of the top 30 differentially expressed genes was generated after row-wise z-score normalization to highlight relative expression patterns across samples. The heatmap reveals a clear separation between breast cancer and normal samples, supporting the results of the differential expression analysis.

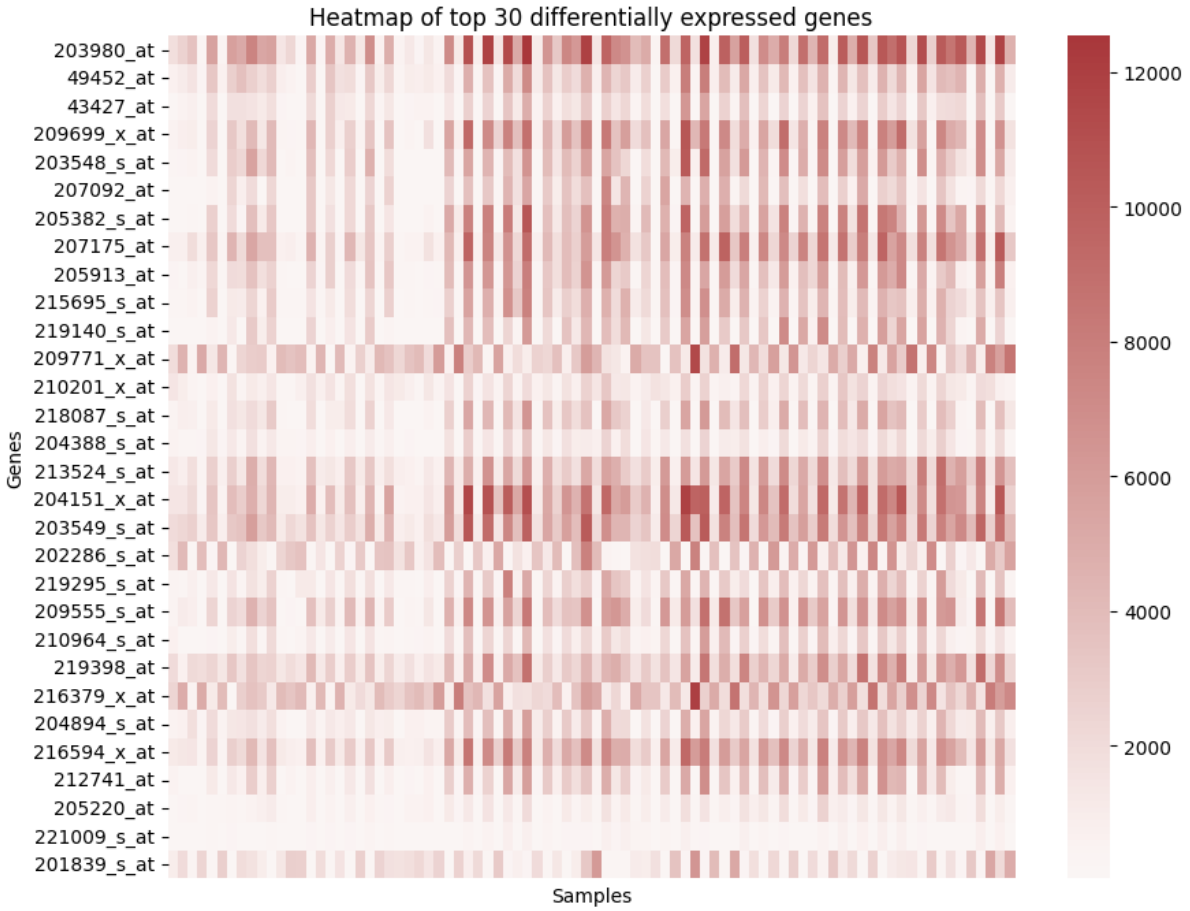


Figure 2: Heatmap of the top 30 differentially expressed genes.

Functional enrichment results were visualized using bar plots showing the top 10 enriched GO Biological Processes and KEGG pathways, ranked according to Benjamini–Hochberg adjusted p-values. These plots highlight the most significantly enriched functional categories associated with breast cancer.

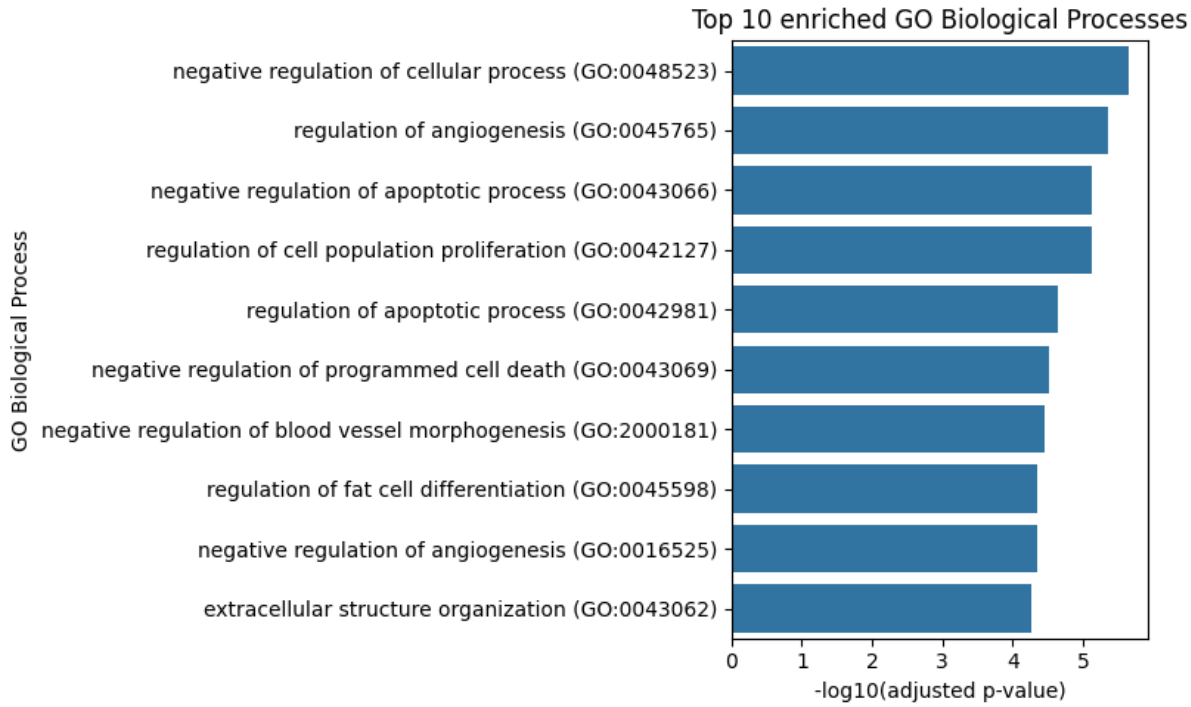


Figure 3: Top 10 enriched GO Biological Processes among the differentially expressed genes. Bars represent the $-\log_{10}$ Benjamini–Hochberg adjusted p-values.

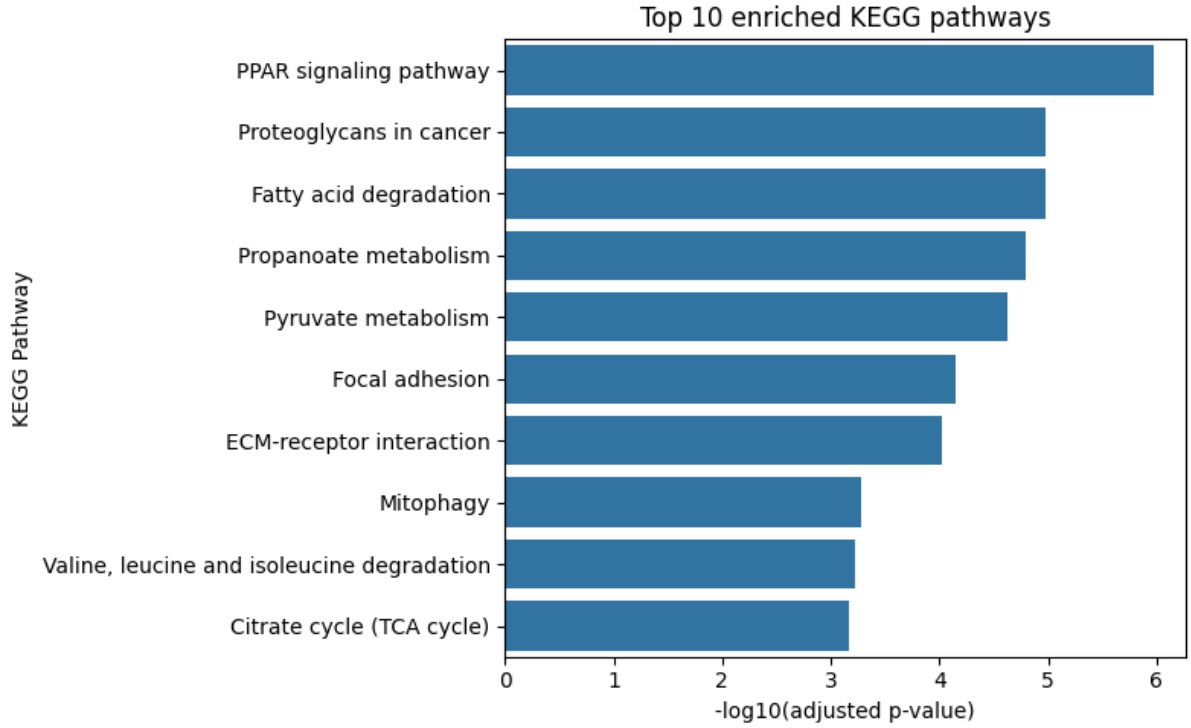


Figure 4: Top 10 enriched KEGG pathways identified from the differentially expressed genes. Bars represent the $-\log_{10}$ Benjamini–Hochberg adjusted p-values.

7 Conclusion

In this study, an end-to-end gene expression analysis was performed to investigate molecular differences between breast cancer and normal breast tissue samples. Differential expression analysis identified 1,667 significantly differentially expressed genes after correction for multiple testing, highlighting substantial transcriptional changes associated with breast cancer. Functional enrichment analysis revealed that these genes are significantly involved in biological processes related to cell proliferation, apoptosis regulation, angiogenesis, and extracellular matrix organization. In addition, KEGG pathway analysis highlighted the enrichment of cancer-related pathways, including focal adhesion, ECM–receptor interaction, and metabolic pathways, which are known to play a key role in tumor progression and metastasis. Overall, the results provide biological insights into the molecular mechanisms underlying breast cancer and demonstrate the effectiveness of enrichment analyses. Future work could include validation of these findings using independent datasets and the application of more advanced statistical models to further characterize breast cancer heterogeneity.